

Teaching Bug Advocacy through Flipped Classroom

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Abstract—Software testing plays a critical role in the development workflow. Nowadays, the significance of teaching software testing principles is recognized to a greater extent than ever before.

The aim of this paper is twofold: (1) to investigate the effectiveness of using a flipped classroom-based context to teach software bug advocacy, and (2) to provide the student's perspective on using flipped classroom to learn how to advocate for a bug. A seminar activity dedicated to bug reports and how to advocate for a bug is the framework for this investigation, with students being split into teams with the aim to perform two major activities: poster creation for one of the strategies from the RIMGEN mnemonics and providing advocacy strategies for a 3 years old bug. The created artifacts and the answers to a questionnaire dedicated to the learning experience are used as tools to analyze and provide answers to research questions.

The results show that flipped classroom-based learning is effective in teaching how to advocate for a software bug. Around 87.75% of students agreed that the poster creation activity helped them better retain the information. A percentage of 61.22% students agreed that time was spent more effectively in class since the text was read outside the classroom, and 82.66% of students also agree that this type of learning provides them with the opportunity to communicate with other students.

Index Terms—Software testing, Bug reports, Software bug advocacy, Flipped classroom, Learning.

I. INTRODUCTION

Teaching software testing [1] is essential because it equips future developers and engineers with the skills to ensure the quality, reliability, and security of software systems. As software becomes increasingly integrated into daily life, from business operations to personal devices, the ability to thoroughly test software [2] is critical to prevent failures, security breaches, and usability issues. By teaching this subject, students learn to identify and resolve bugs early, understand testing strategies like unit, integration, and performance testing, and grasp the importance of quality assurance in the software lifecycle. Mastering testing concepts also fosters a mindset of precision and attention to detail, which are vital for creating robust, user-friendly applications. Educating students in software testing not only improves the quality of the software they will build but also helps meet industry demands for high-quality, secure, and reliable software products.

The Computing Curricula 2020 (CC2020) [3] adopts a competency-based approach to structuring computing disciplines, organizing sub fields around key skills and abilities. *Verification and Validation in Software Engineering* has a level

of minimum 3 to maximum 5 for undergraduate (computing knowledge). The main competences in software testing that are mentioned in the Software Engineering competences refer to performing an integrative test and analysis of software components by using black-box, conducting a regressive test of software components, conducting a test utilizing appropriate testing tools, and planning and conducting processes to design test cases. The new 2023 curricula [4] also emphasizes the importance of *Verification and Validation* and outlines several computer science (CS) core concepts, adding testing objectives, why testing matters, test kinds, usability, reliability and security, and differences between tests and production code.

Engaging and motivating students with the subject of software testing can be particularly challenging, especially when it is taught in the final semester of their studies. Many students struggle to see the real-life relevance and impact of testing in software development, making the topic appear less important or exciting. Without a clear understanding of how critical software testing is in preventing bugs, improving software quality, and ensuring user satisfaction, students may undervalue its importance. Educators must therefore emphasize the practical significance of testing in professional settings, demonstrate real-world cases where inadequate testing led to significant issues, and incorporate hands-on activities [5], [6], [7] to make the subject more tangible and engaging. Highlighting how testing is essential to delivering reliable, secure, and efficient software can help shift student perceptions and foster a deeper appreciation for the discipline [8].

Bug Advocacy [9] is a critical aspect of software testing [10], emphasizing the effective communication of bugs to ensure they are understood and prioritized by stakeholders like developers and project managers. By presenting bugs with clear evidence of their impact on system functionality and user experience, testers advocate for their resolution, enhancing software quality. The RIMGEN method [11], which stands for *Replicate, Isolate, Maximize, Generalize, and Externalize*, supports this advocacy by guiding testers through a structured process. This method ensures bugs are reproducible, well-understood, and their broader impact clearly stated, increasing the chances of timely fixes. Additionally, incorporating insights from human factors in computing, which explores how people interact with technology, strengthens Bug Advocacy [12]. By linking bugs to real-world user challenges, one can better advocate for fixes on certain issues, ensuring that both technical

and human aspects of software are addressed in development. This human-centered approach not only improves the software's technical performance but also enhances user satisfaction and business outcomes.

In teaching software testing, successful learning involves using a mix of active and engaging techniques [5], moving past standard lectures to enhance both theoretical understanding and practical abilities. Interactive approaches help maintain students' focus and engagement, making learning more effective and satisfying. A harmonious blend of theory with practical application fosters critical thinking, teamwork, and problem-solving abilities, equipping students for the challenges of real-world scenarios.

The aim of this paper is twofold: first, to investigate the effectiveness of using a flipped classroom-based context to teach bug reports and bug report advocacy, and second, to provide the student's perspective on using flipped classroom to learn how to advocate for a software bug.

The research investigation is formulated using two research questions:

- RQ1: Is flipped classroom effective in teaching software bug advocacy? The results show that flipped classroom-based learning is effective in teaching bug report advocacy, the artifacts created by the students during the poster-creating activity and during the *3 years old bug* problem being an incontestable proof of it.
- RQ2: What are student perceptions regarding flipped classroom-based software bug advocacy learning activities? Approximately 71.43% of students held a favorable view of the flipped classroom, indicating that this approach to learning aided their comprehension of the concepts. A percentage of 61.22% of students felt that classroom time was used more efficiently because the text was read outside of class, and 82.66% concurred that this learning approach facilitates communication with peers.

The contributions of this research investigation are:

- **A new flipped classroom-based pedagogical design for teaching software bug report advocacy concepts:** A flipped classroom-based activity is designed: students received before the activity information regarding various aspects of bugs, bug reports, and strategies of how to advocate for a bug, while during in-class activity students created posters based on the home study material, and then applied them to elaborate on strategies how to advocate of a *3 years old bug* problem.
- **Evaluation through the seminar by the created artifacts and a questionnaire:** The effectiveness of the flipped classroom-based learning approach is evaluated based on the artifacts created by the students during in-class activities that used the learned concepts (poster creating using the learned concepts before the seminar and providing strategies of how to advocate for a bug). A questionnaire provides opinions of the students regarding the flipped classroom-based learning approach.

The rest of the paper is organized as follows: Section II briefly presents the concepts to be learned (bug report and bug advocacy), the learning methods (flipped classroom), and the related work analysis. The methodology (research questions, seminar details, activities) is outlined in Section III. The results are presented in Section IV, emphasizing the students perception and learning outcome, along with the answers to the research questions. Section V provides threats to validity. Section VI states the outcome of our experience.

II. BACKGROUND

In this section, we provide the background on the software testing concepts that we analyze, namely, bug reports and bug advocacy, and on the learning methodology, flipped classroom. The section ends with a related work analysis.

A. Bug Reports

A bug report [2] is a detailed document created by testers or users to describe an issue or flaw in a software system. It serves as a key communication tool between testers and developers, outlining problems encountered during testing. A well-constructed bug report typically includes several key elements: a unique identifier or bug ID, a summary or title of the issue, a detailed description of the bug's behavior, steps to reproduce the issue, the expected versus actual result, and most importantly severity or priority levels to help understand the urgency.

The importance of writing a well-structured bug report cannot be overstated, as it directly affects the efficiency of the software development process. A poorly written bug report can lead to misunderstandings, wasted time, and unresolved issues, slowing down development and impacting the software's overall quality. On the other hand, a clear, concise, and well-documented bug report improves communication between testers and developers, allowing for quicker identification and resolution of the issue. This not only speeds up the debugging process but also ensures that critical bugs, which might affect the software's performance or user experience, are prioritized and addressed. Furthermore, well-written bug reports contribute to the creation of robust and reliable software, ultimately improving user satisfaction and minimizing costs associated with fixing defects late in the development cycle.

A well-crafted report not only helps developers quickly identify and address the issue but also saves time by minimizing the need for follow-up questions or clarifications. Furthermore, clear and detailed bug reports foster stronger collaboration and communication between testers, developers, and stakeholders, leading to a more efficient workflow and a better working relationship across teams.

Writing well-structured bug reports is crucial as it directly impacts the efficiency and success of the software development process [13]. An important step on writing good bug reports is advocating for them to be fixed.

RIMGEN [9] suggests an approach to bug reporting that helps improve bug advocacy by guiding testers through a set of steps as follow:

- **Replicate:** Ensure the bug can be reproduced consistently.
- **Isolate:** Identify the specific conditions that trigger the bug.
- **Maximize:** Determine the most severe impact of the bug.
- **Generalize:** Check if similar issues appear elsewhere in the system.
- **Externalize:** Clearly communicate the bug's significance to stakeholders.

Regarding RIMGEN [9], there are different degrees of using one strategy of the mnemonic based on the type of the investigated errors, namely, design or coding errors. For example, is *essential* to use *Replicate* for a coding error and is *rarely important* in case of a design error. However, is *essential* for a design error to use *Externalize* but for a coding error is *sometimes useful*. Cem Kaner [9] highlights that *Neutral tone* is *essential* when advocating both types of bugs. This method promotes the creation of clear, persuasive reports, increasing the likelihood of timely and effective bug resolution.

B. Learning Methodology

The flipped classroom model [14] is transforming education by inverting conventional teaching approaches. In this model, students first interact with new content outside the classroom through various media, including videos, podcasts, articles, and virtual reality. This method promotes self-paced learning and allows classroom time to be utilized for engaging, hands-on activities that enhance comprehension.

The flipped classroom should always enhance the in-class experience, ensuring that class time is dedicated to interactive and engaging activities. By shifting content delivery outside the classroom, this model allows for a more dynamic learning environment where students can actively participate, collaborate, and deepen their understanding through practical application as per also stated in [15]. This also requires some hands-on practice, which is helpful.

Although a variety of subjects have implemented the flipped classroom model, its primary application has been in teaching foreign languages such as [15] and [16]. Other disciplines also use the flipped model, though they are less common in comparison. This approach is suited for most of the subjects that also have a practice base, as it supports enhancing skills like speaking and listening as it also stated by [14], and also facilitating communication among students.

The flipped classroom model is particularly effective for computer science students, as it allows them to independently learn core concepts, like programming and theory, before class where it is usual to work on hands on problems. This enables classroom sessions to focus on coding, problem-solving, and debugging. During this process students can apply and deepen their understanding on new concepts and apply also on existing projects. The main advantage is that instructors can provide instant feedback on challenging topics, promoting active learning and helping students develop practical skills essential for real-world programming tasks. This interactive, application-focused approach better prepares them for the high speed environment the industry has to offer.

C. Related Work

Flipped classroom models significantly enhance students' academic achievement as revealed by several studies on different subjects [15] and [16]. By actively engaging students in the learning process, this approach fosters the development of critical skills such as analytical thinking, problem-solving, creative thinking, and meta-cognitive awareness. These skills are integral to achieving academic success, as they enable students to approach complex problems systematically, think innovatively, and reflect on their learning strategies.

Several flipped classroom approaches in the context of computer science are outlined next. The experiences of flipping four different computer science classes are described in [17]: how videos were created or sourced, how active learning activities were designed for in-class (pair programming labs, group problem-solving activities, and flexible quiz activities), and how the teams were structured. Another study [18] showed that the flipped classroom instructional model improved the students' perception of the learning experience and students' dedication and engagement in the computer science degree. Initially, instructors prepare a series of videos containing the lecture material they intended to cover. At the start of the class, students took a brief quiz regarding the videos they have already viewed. This quiz was structured to assess their proper comprehension of the concepts discussed in the videos. Once students have completed the quizzes, the instructor provided feedback on their scores. These activities were confined to a quarter of the total lecture duration. The remaining class time was devoted to activities that promote student engagement and deepen their understanding of the concepts previously introduced. This is achieved through exercises, in-class labs, and other hands-on activities. The objective was to optimize classroom time to consolidate the learning process by applying what students have learned before the lecture. Some instructors have incorporated additional active teaching methods within the classroom, such as team-based learning. The research presented in [19] concluded that the flipped classroom approach is beneficial for students as it fosters greater self-directed learning (independent learning). Their learning styles could be determined, enabling the creation of participatory environments with teachers serving as facilitators. Flipped classrooms facilitate active learning and engagement, emphasizing hands-on experiences over rote memorization (a conventional approach), supported by innovative technologies utilized in higher education.

A thorough review undertaken for this study revealed an absence of research specifically utilizing the flipped classroom model to enhance students' understanding of bug advocacy. Our research investigates the use of flipped classroom approach for teaching bug advocacy in the context of computer science program. As in the above studies, the students were provided with various resources to study the concepts to be learned before the class, and in-class activities consisted of activities where students created various artifacts based on the already

available resources, and also applied the learned concepts to solve a practical problem.

III. METHODOLOGY

The objective of this research is to investigate the use of flipped classroom for learning software testing concepts, how to write a good bug report (how to advocate for a bug, or bug advocacy). Figure 1 depicts the elements of our investigation: software testing concepts (bug reports, RIMGEN), learning methodology (flipped classroom).

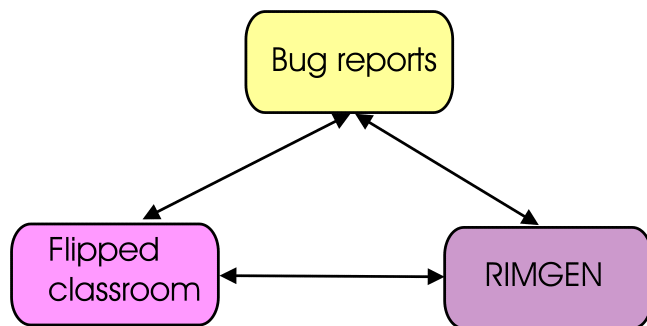


Fig. 1. Research design overview.

The activity, conducted in one of the seminars, consisted of two major parts: (1) creation of a A3 poster for one of the RIMGEN perspectives, (2) describing two ways to advocate for a bug (under a specific context). Details about the activity will be provided in Section III-C. The students were given, before the seminar, various resources to document the writing of a good bug report: text documents, video lectures, online discussions, conference discussions.

A. Research Question

The objective of this study is next formulated into several research questions that target various aspects of learning.

RQ1: Is flipped classroom effective in teaching software bug advocacy?

RQ2: What are student perceptions regarding flipped classroom-based software bug advocacy learning activities?

The direct observations on the students' feedback on the seminar activity, discussion during the activity and the analysis of the team results are used to answer RQ1. The feedback on each activity (poster and bug report creation)-based learning and a survey that the students answered were used to answer RQ2.

B. Course and Participants

The course under analysis is set up as an undergraduate level, i.e. in the "Computer Science" program (third year of study) at Babes-Bolyai University, with the title *Software*

Systems Verification and Validation. The course spans 12 weeks, comprising 2 hours of lectures, 1 hour for seminars, and 1 hour for lab sessions per week, alongside allocated time for home study (project hours). The verification and validation concepts covered include: inspection and review, designing test cases with black-box and white-box methods, various integration strategies for different testing levels, exploratory testing, UI web testing, symbolic execution, model checking, and correctness. The seminar and laboratory sessions focus on various aspects of the concepts taught; this paper, however, will only discuss the one we are analyzing (one seminar with a specific topic and learning methodology, namely bug advocacy using flipped classroom).

C. Description of Activities

We provide next the designed activities for the seminar. The allocated time for the seminar was structured in two major parts: (1) poster creation, each team constructing a poster with information for one of the letter of the mnemonic RIMGEN, (2) each team provided two ways to advocate for a specific bug with a context given. Figure 2 graphically depicts the time dedicated to each part of the seminar, incorporating the results discussion and debriefing, along with the time to fill-in the questionnaire about the activity.

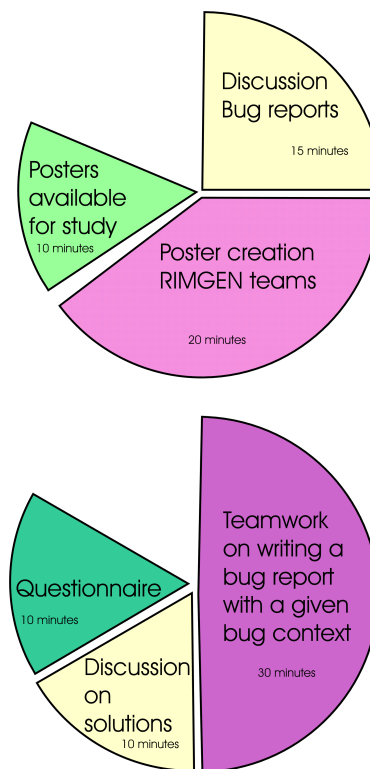


Fig. 2. Timeline of the flipped classroom-based bug advocacy learning activity.

The details on the two activities (poster and bug advocacy problem) are provided in Figure 3. Each team received one of the letter of the mnemonic RIMGEN and created a poster with information specified in the Figure 3 (upper part of the figure). The same team, after the posters were available

for study, received a bug advocacy problem with the aim to propose strategies/solutions to advocate for that specific bug (and context), with description on the lower part of Figure 3.

RIMGEN poster

What it is? When it is used? Why it's useful?
Description/Definition
Coding error vs. Design error
Essential/Important/Useful

Example for:
Coding error
Design error

Bug context Problem: A three year old bug.
A bug has been in a product for 3 years and the project manager refuses to fix it.

- Describe two ways that you could check whether this bug has customer impact.
- In each case, include in your description:
 - What kind of information you are looking for (be specific).
 - How you would get that type of information.
 - How that information would be useful in demonstrating the magnitude of customer impact.

Fig. 3. The description of the two activities: (1) poster, (2) bug advocacy problem

The activities, poster creation and bug advocacy problem, were performed during seminar hours with 7 groups of around 30 students each. For each seminar, 6 teams were created that first constructed the poster for one specific letter from the mnemonic RIMGEN and included information as specified in Figure 3. The same teams, in each seminar, discussed and provided solutions to advocate (or not) for the bug to be fixed or not. The groups were formed by the students based on their positions at the desks and if needed, some students moved to other teams where there were fewer members such that the teams were balanced in terms of number of members.

Prior to the seminar, the students received information from where to document themselves regarding bug advocacy and RIMGEN, as follows:

- Cem Kaner on Bug Advocacy, [9].
- BBST Bug Advocacy Lecture 6: Writing Clear Bug Reports, [12].
- EXTRA Pdf document from [11].
- RIMGEN mnemonic EXPLAINED on Testival 2016, the video was provided to the students in Teams, [20].
- Altom Cards, [21].
- Other resources, How to Write a Good Bug Report (and be a Bug Advocate), [22].

- Other resources, How to Write a Good Bug Report, use RIMGEN, [23].

D. Creating Questions Reflecting Bug Advocacy concepts

An online survey entitled *RIMGEN Flipped Classroom* was provided at the end of the seminar. Table I provides several of the survey perspectives related to the flipped classroom teaching methodology used to learn about bug advocacy, along with perceptions of students related to learning. The questionnaire's questions are available in figshare [24].

TABLE I
STUDENT SURVEY QUESTIONS DESCRIPTION.

Perspective	Questions's Target Aspects
Previous knowledge	Individual/group study before seminar, lecture with bug report concept.
Flipped classroom	Comparison with traditional class instruction, benefit in future education, opportunity to communicate, time in class more effective, better understanding.
Seminar activity	Poster, bug advocacy problem, debriefing.

The number of students completing the survey was 98, all of them participated in the seminar work activity in the 6 teams described as above. Several information regarding characteristics of the respondents are provided in Table II.

TABLE II
SUYEV RESPONDENTS DATA INFORMATION.

Information	Number
Number of respondents	98
Female	35
Male	62
Other	1
Age	20 (3), 21 (66), 22 (26), 24 (3)

IV. ANALYSIS OF RESULTS

In this section, we analyze the results, beginning with a discussion on the solutions offered by the teams, and then interpreting the questionnaire responses. The latter part of the section addresses the research questions.

A. Results Discussions

1) *Activity Results:* We outline in what follows the results of the seminar teams regarding the poster and bug advocacy problem.

In Figure 4 we present one sample for each letter perspective from the mnemonic RIMGEN that the students created in their posters following the information requested in Figure 3.

Students used the information they had to prepare before starting the seminar (as described in Section III-C), the information being also available for them during the seminar (in printed format or online). Students created structured posters, following the questions asked to be answered in Figure 3, using various colors and drawn elements such as arrows, circles, images, suggestive icons, and even their own drawings. When studying and analyzing the artifacts (posters) created by students, those in Figure 4 and the others also, we noticed that

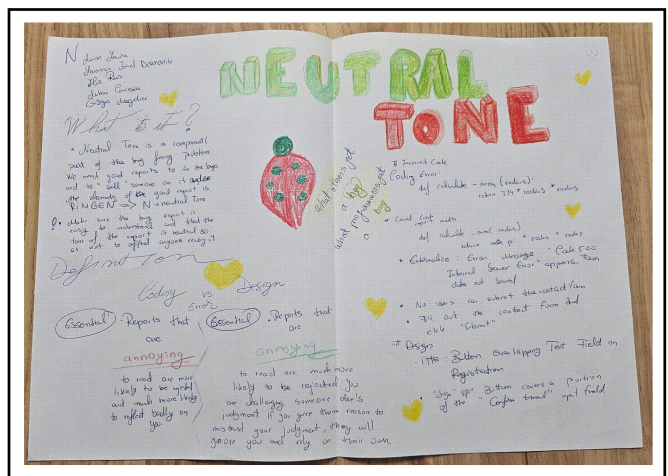
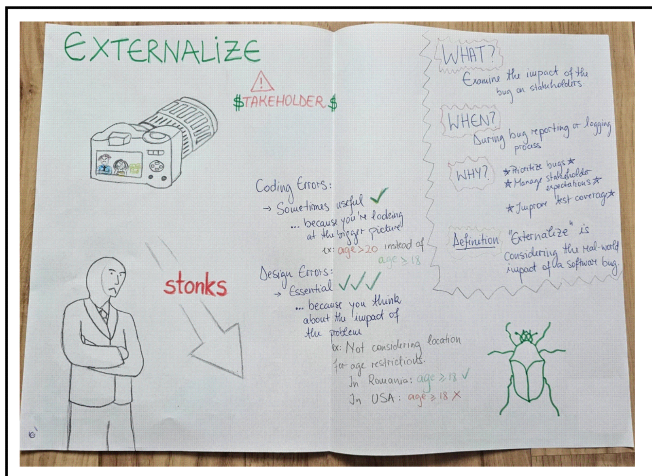
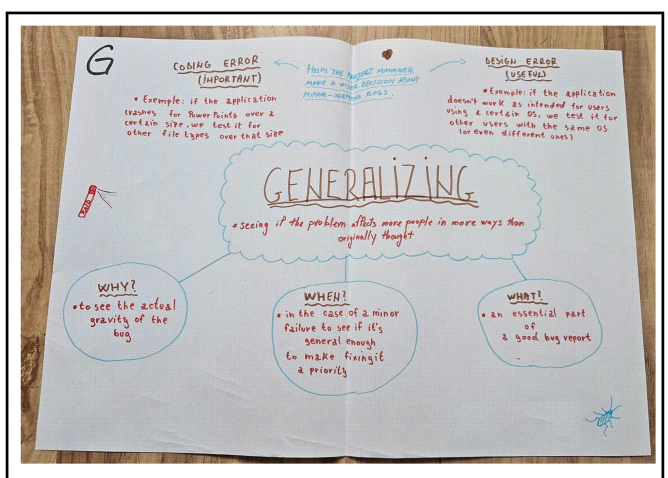
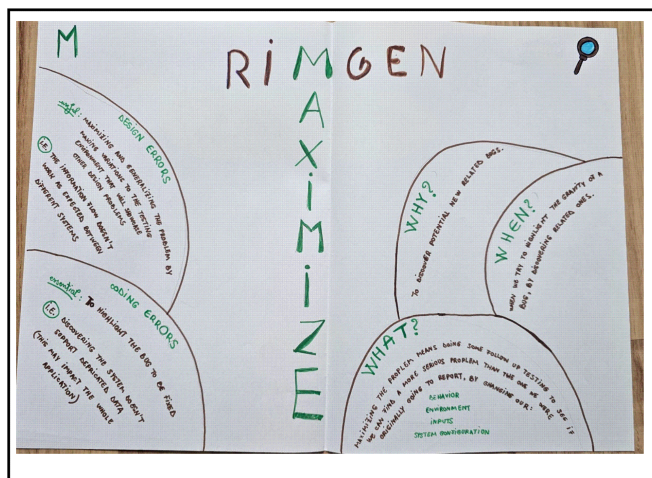
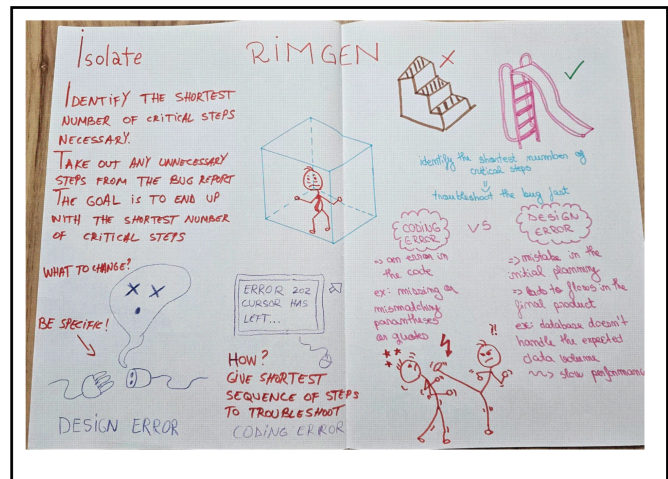
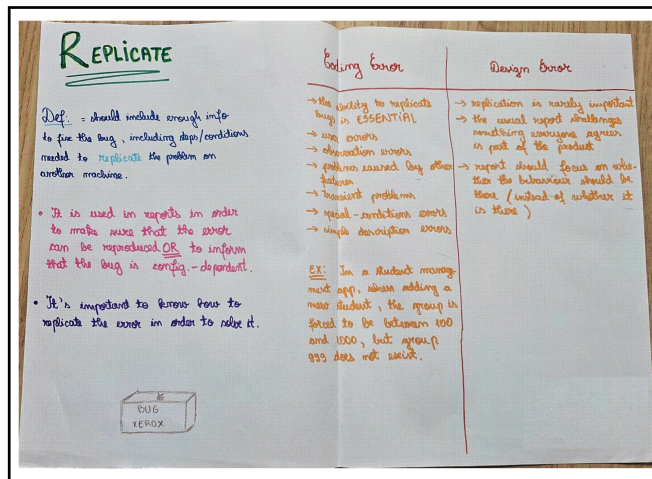


Fig. 4. Sample posters created by students for each letter perspective from the mnemonic RIMGEN.

students followed the guideline regarding what information the posters should contain, they structure the information in the poster based on them. The students provided the main idea of the specific mnemonic RIMGEN letter they studied, they described the errors from the two different perspectives (design and coding) with examples, and also displayed the information on why, when, and what that specific strategy is used. Working in teams also helped them to finalize the posters in time with the requested information.

In Figure 5 we have plot the used of RIMGEN perspectives by each poster team in order to solve the *3 years old bug* problem, for example, the team that created the Maximize poster used in principal the Maximize (in three posters) and Externalize (in three posters) strategies and also Generalize (in two posters), Replicate (in two posters) and Isolate (in one poster). The data for a column in the figure uses the data from all posters created in all seminars by that specific mnemonic letter strategy, thus from seven posters. It can be noted that all teams used the Generalize approach, followed by the Externalize, Maximize and Replicate perspectives. Some of the teams used also the Isolate and Neutral tone strategies. The teams that used almost of the perspectives were the teams that created the posters for the Generalize and Maximize perspectives.

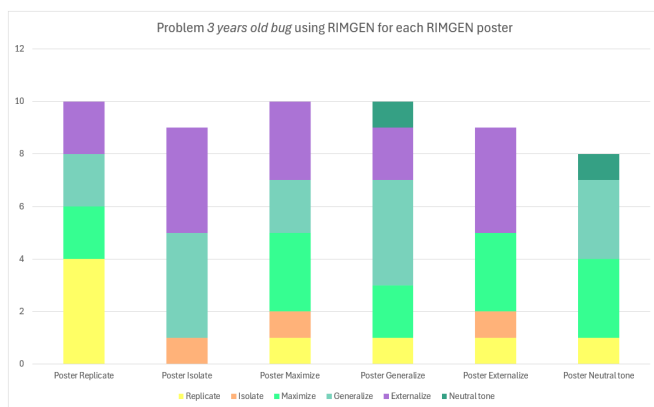


Fig. 5. The use of RIMGEN elements for the *3 years old problem* by each poster team.

2) *Questionnaires Results:* We outline in what follows the answers to the questionnaire questions.

Student's prior study (or knowledge) about bug reports and RIMGEN is outlined in Figure 6. Regarding the student's prior knowledge on bug reports and RIMGEN (watched the videos, read the information provided in the links) before the seminar activity, there is a half-and-half distribution of the students that have (51.02%) and did not have (48.98%) informed about the topics. Regarding participation on the second lecture that contained information about testing and bug reports, only 37.76% were physically present in the classroom, while during lecture 9, only 9.18% of students were present.

Student's opinions on the flipped classroom learning about bug reports and RIMGEN are outlined in Figure 7 and Figure 8. From Figure 7 we can notice that around 76.53%

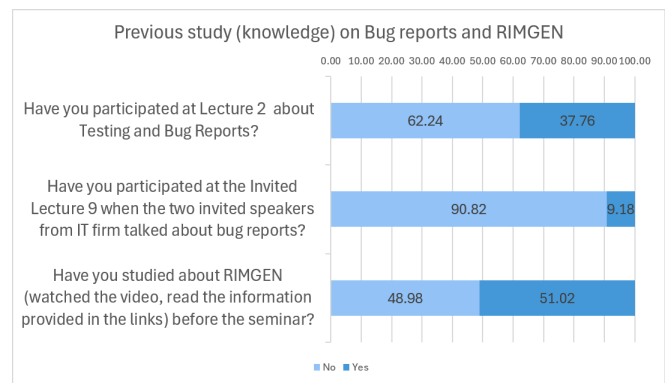


Fig. 6. Student prior study (or knowledge) about bug reports and RIMGEN.

students agree that flipped classroom are more engaging than traditional classroom instruction, 19.39% have a neutral opinion, and around 4% disagree. Regarding traditional versus video presentation type of lessons, the opinions are balanced: 28.57% prefer traditional, 35.71% are neutral, and 28.57% prefer video lectures. Students agree in a large percentage, 69.39%, that learning using flipped classroom will benefit them in their future education.

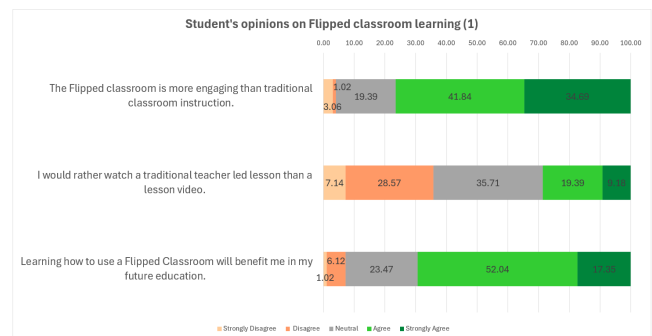


Fig. 7. Student's opinions on flipped classroom learning about bug reports and RIMGEN (1).

A percentage of 71.43% students (in Figure 8) agree that flipped classroom helped them to understand better the concepts, 25.51% had neutral opinion, and 3.06% disagree. Regarding the efficiency of the time spent in classroom considering that text was read outside the classroom, 61.22% of students agree with it, 32.65% were neutral, and 6.12% disagree. The majority of students agree that flipped classroom provided them the opportunity to communicate with other students, 82.66%, only 4.08% disagree and 13.27% had a neutral opinion.

Student's opinions on the bug reports learning (poster creation, RIMGEN, bug advocacy problem, debriefing) is outlined in Figure 9. The opinions of the students regarding the activities during the seminar are outlined: the poster creation activity on the RIMGEN elements is perceived as positive by the students, 87.75% of students agree that this activity helped them to better retain the information, 72.45% students agree that having all the posters available helped them to think about solutions for the "3 year old bug", and 79.59% of students

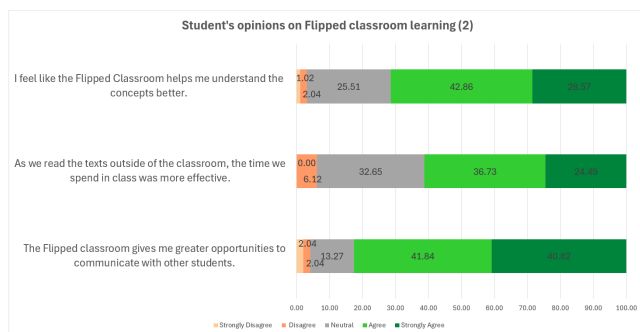


Fig. 8. Student's opinions on flipped classroom learning about bug reports and RIMGEN (2).

agree that this “3 years old bug” provided them a deeper understanding of RIMGEN and how to use it.

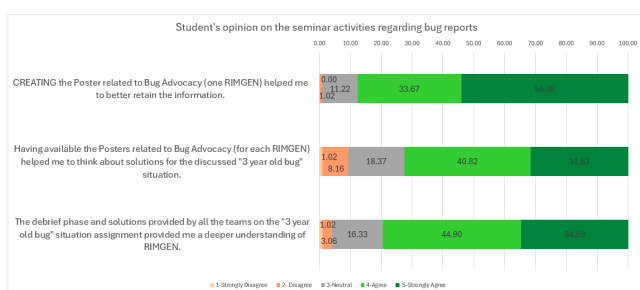


Fig. 9. Student's opinions on the bug reports learning (poster creation, RIMGEN, bug advocacy problem, debriefing).

B. Answers to Research Questions

In this section, we revisit the research questions and, drawing from the prior results analysis, offer responses to them.

RQ1: Is flipped classroom effective in teaching software bug advocacy? To formulate the answer to this question, we use the responses of the questionnaires provided to the students during the seminar activity along with the artifacts created by the students for the two activities during the seminar, the posters, and the 3 years old bug problem.

Answer RQ1: Flipped classroom is effective in teaching software bug advocacy. More than half the students informed themselves before the learning activity from the documents that were provided before the activity. All students actively participated in the poster activity, elaborating detailed and rigorous posters regarding RIMGEN-based bug advocacy strategies. Around 87.75% of students agreed that the poster creation activity helped them to better retain the information. Moreover, for the 3 years old bug problem, students mostly used the Generalize and Maximize perspectives, showing a good understanding of how to use the mnemonic.

RQ2: What are student perceptions regarding flipped classroom-based software bug advocacy learning activities? To answer this question we have used the responses of the students to the questionnaire provided at the end of the activity, questions that were specifically constructed regarding student's perception on the flipped classroom and bug advocacy learning.

Answer RQ2: The student's perceptions regarding the flipped classroom were positive, around 71.43% of students agree that this way of learning helped them to understand better the concepts, 61.22% of students agreed that time was spent more effectively in class since the text was read outside the classroom, and 82.66% of students also agree that this type of learning provides them with the opportunity to communicate with other students.

C. Implications and Recommendations for Teachers

This section provides implications and recommendations for teachers regarding using the flipped classroom to teach software testing, more specifically how to advocate for a bug and how to construct a clear bug report.

Implications for teaching bug advocacy using flipped classroom

Next, we outline several implications of teaching bug advocacy using flipped classroom. First, the pre-class preparation for contextual learning is enhanced because the students review materials before the class. They arrive in the class with a foundational understanding, enabling deeper exploration during the class. Secondly, the class time focuses on interactive activities, fostering group discussions on challenging bug and how to advocate for that bug in the bug report, collaborative learning allowing students to learn from each other. Thirdly, the teacher provides immediate feedback on the students' bug reports allowing students to reflect on their strengths and weaknesses and apply improvements. Thus, students build competence in articulating the value of bug fixes, ensuring that students internalize the practices of how to advocate for a bug.

Recommendations for teachers

For the pre-activity preparation it is important to provide students several types of instruction materials (readings, video tutorial, RIMGEN descriptions, case studies, bug report examples, etc and also a list of guiding questions). Another recommendation would be to provide students also with interactive quizzes or discussion forums to encourage engagement before the seminar.

Regarding the activity in the classroom, a good practice is to divide the time during the activity in hands-on activities that allow remembering the studied information before the class, along with activities that simulate real-world bug advocacy scenarios. The use of RIMGEN (or similar tools) to gather relevant data and metrics to support their advocacy argument is also recommended.

An important stage in the learning process is the reflection part, thus providing students the means to reflect on their experience by documenting the steps they took to advocate

for the bug, challenges they faced, and lessons learned will make the new assimilated information to stick in their long time memory [25]. Peer feedback is also encouraged, focusing on clarity and impact of the created bug report.

V. THREATS TO VALIDITY

We discuss possible challenges to our research validity in this section and detail the strategies we employed to tackle and reduce these risks.

Internal Validity. A cohort of students from a particular software testing class participates in the study. If these students are more predisposed to, or have prior experience with, flipped classroom learning methods, there could be a potential selection bias. If individuals are inclined towards advocating bugs or prefer learning by creating posters, these findings may not be broadly applicable to all students involved in software testing. Furthermore, if students possess previous knowledge about writing bug reports or advocating for bugs, as well as experience with flipped classroom exercises from earlier terms, this familiarity might impact their performance, potentially resulting in exaggerated outcomes.

External Validity. External validity denotes the degree to which a study's results can be applied to different contexts. The research is restricted to a particular group of students at a single academic institution, leading to uncertainties about the generalizability of the findings to different student demographics. Moreover, the seminars that incorporate flipped classroom activities might not completely emulate actual software engineering settings.

Construct Validity. Utilizing a single approach, such as surveys or observational feedback, to evaluate the effectiveness of a flipped classroom activity to advocate for a software bug may restrict the depth of the assessment. Incorporating a variety of approaches, including qualitative interviews, quizzes, and comparative analysis with traditional non-flipped classroom teaching methods, would enhance the study. If key ideas like "collaboration" or "bug advocacy testing skills" are not clearly outlined and uniformly evaluated among participants, it can create uncertainty about what is being evaluated. This might result in a misunderstanding of the outcomes.

VI. CONCLUSIONS AND FUTURE WORK

As software systems grow in complexity, testing has become an essential part of the development process. The importance of teaching software testing principles is now recognized more than ever before.

This paper has a dual objective: first, to explore how effective a flipped classroom approach is for teaching software bug advocacy, and second, to present the student's viewpoint on utilizing a flipped classroom for learning bug advocacy.

This investigation is centered around a seminar activity focused on bug reports and methods for effectively advocating for bugs. In this setup, students are divided into teams tasked with two main objectives: designing a poster for one strategy from the RIMGEN mnemonics and formulating advocacy strategies for a bug that has persisted for three years. The

artifacts produced, along with responses from a questionnaire about the learning experience, served as instruments for analyzing and addressing the research questions.

The findings indicate that employing a flipped classroom approach is beneficial for teaching the advocacy of software bugs. Approximately 87.75% of students concurred that the activity of creating posters aided in improving their information retention. Additionally, 61.22% of students believed that class time was utilized more efficiently given that reading was done outside of class, and 82.66% of students agreed that this learning method offers a chance to interact with their peers.

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